

Handout 5.4: Stain Station Record

Name: _____ Date: _____ Period: _____ Station: _____

Directions: Three stains. Three treatments on each. Same method every time so the comparison is fair. Fill in the table as you go, then finish the cost line and the conclusion.

Station rules: gloves if you want them, and nobody has to say why. No treatment goes near a face. The rubbing alcohol cup stays on the tray. Used swabs go in the trash, not back in the water bowl. Wash hands at the end.

The method, the same for every test

1. Read the stain name and the treatment number.
2. Put the treatment on with a cotton swab or the toothbrush.
3. Work it for 30 seconds. Same pressure every time. Watch the clock, do not guess.
4. Rinse the spot in the clean water bowl.
5. Blot with a paper towel. Do not rub it dry.
6. Rate the result 0 to 3 and write it in the table.
7. Hang the swatch on the line with a clip, letter side out.

The rating scale: 0 = no change. 1 = a little lighter. 2 = mostly gone. 3 = gone.

The three treatments

Number	What it is
1	Cold water only
2	One drop of dish soap in cold water
3	The home remedy for that stain: baking soda paste for grease, white vinegar and water for the drink, rubbing alcohol for the ink

Part 1: The results

These stains were made yesterday on purpose. A fresh stain comes out with water, and that is not the problem people actually have at home.

Stain	Treatment 1 (cold water)	Treatment 2 (dish soap)	Treatment 3 (home remedy)	Which treatment won
Grease (cooking oil)				
A drink (grape juice)				



Ink (ballpoint pen)				
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Part 2: What I noticed

2a. Which stain was hardest to move at all? _____

2b. Did any treatment make a stain worse or spread it? Which one, and what happened?



2c. Why does grease need something different from what ink needs? Your best guess.

Part 3: The cost line

Pick one garment the stain could have been on. Use the prices on the slide or your own.

The garment	What it costs to replace	What the treatment cost	The difference (what you saved)
	\$	\$	\$

One sentence: was the 30 seconds of treatment worth it? Say why with the numbers.

Part 4: The class pattern

Fill this in during the debrief, from all three stations.

Stain	The treatment that won across the class	Why that one works (in your own words)
Grease		
A drink		
Ink		

Finish the sentence: Every garment saved from the trash is one you did not have to

and one that did not go to _____.

Teacher key

Part 1, the expected pattern. Ratings will vary by fabric and by how set the stain is. The pattern is what matters, not the exact numbers.

Stain	Cold water	Dish soap	Home remedy	The winner
Grease (cooking oil)	0 to 1. Water and oil do not mix, so plain water does almost nothing	2 to 3. Dish soap is made to break up oil, which is exactly this job	1 to 2. Baking soda paste absorbs some oil and helps, especially on a fresh stain	Dish soap
A drink (grape juice)	1 to 2. A sugary drink stain is water based, so cold water moves a lot of it	2. Soap helps some	2 to 3. The acid in white vinegar breaks up the dye	Home remedy, with cold water a close second
Ink (ballpoint)	0. Ballpoint ink is not water soluble	0 to 1	2 to 3. Rubbing alcohol dissolves the ink's carrier	Rubbing alcohol

Part 2. 2b: the honest answer is yes, ink spreads when it is rubbed with water or soap, which is why blotting matters more than scrubbing; that is a real finding and it earns full credit. 2c: grease is oil, and oil needs something that breaks up oil (a detergent); ink is dissolved in something that is not water, so it needs a solvent like alcohol. A student who says “different stains are made of different stuff” has the idea.

Part 3. Any garment and any reasonable prices. A Meets answer names a real replacement price, a treatment cost that is a fraction of a cent to a few cents, and a one-sentence conclusion that uses both numbers. Sample: a dish soap bottle at \$4 does about 200 stains, so one treatment costs about two cents against a \$12 shirt.

Part 4. The class pattern is the three winners above. The closing sentence: you did not have to buy, and it did not go to a landfill. That is the plain-language version of CTE SUS 1. Resources e), practices that promote stewardship of environmental resources.

Setup reminders for the teacher. - Make the stains the day before. This is the one setup step the lesson depends on. - One cup of rubbing alcohol per station, on a tray, counted out and counted back in at the end, the same way needles get counted starting in Lesson 5.5. - Pull the Safety Data Sheets for the rubbing alcohol and the white vinegar and know where they are before a student asks. - Score the record complete, partial, or missing: complete means nine ratings filled, a winner picked for each stain, and the cost line finished.

Standards checked by this handout: NYS ATPD 3. Maintenance a); CTE SUS 1. Resources e); NYS FACS Standard 2 (Intermediate) PI C; CDOS 3a Basic Skills; National FCS 3.0 16.4.3. The activity design comes from the Sustainability theme module's own illustrative activity



for the Apparel and Textile Production and Design content module, “Stain Removal Comparison.”